

hourglass space

YoonKi Kim

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Abstract

$$\mathbf{Goal:} \quad H_{R,\rho}^{\text{phys}}(t) \geq \sigma R I, \quad E_0^{\text{phys}}(R, t) \geq \sigma R.$$

$$\text{Here} \quad \sigma = \frac{g^2 c_{\text{geo}} C_2(\rho)}{2A_*} > 0.$$

The charged sources are implemented as finite-dimensional endpoint representation factors,
not as singular point-source insertions in the gauge-field sector.

Consequently no source self-energy counterterm appears, $\kappa_\Lambda = \kappa = 0$.

1. Hourglass geometry.

$$T_* > 0, \quad \rho : [-T_*, T_*] \rightarrow (0, \infty), \quad \rho(-t) = \rho(t), \quad \rho \text{ Lipschitz}, \quad \rho \text{ nondecreasing on } [0, T_*].$$

$$\Omega_{\text{hg}} := \{(t, x) \in \mathbb{R} \times \mathbb{R}^3 : |t| < T_*, \ |x| < \rho(|t|)\}.$$

$$\Omega_+ := \Omega_{\text{hg}} \cap \{t > 0\}, \quad \Omega_- := \Omega_{\text{hg}} \cap \{t < 0\}, \quad \Sigma_0 := \Omega_{\text{hg}} \cap \{t = 0\}.$$

$$\tau : \Omega_{\text{hg}} \rightarrow \Omega_{\text{hg}}, \quad \tau(t, x) = (-t, x).$$

$$\tau^2 = \text{Id}, \quad \tau(\Omega_+) = \Omega_-, \quad \tau(\Omega_-) = \Omega_+, \quad \tau(\Sigma_0) = \Sigma_0.$$

2. Forward and backward time flows.

$$\Phi_u^+ : \Omega_+ \supset D_u^+ \rightarrow \Omega_+, \quad \Phi_u^+(t, x) = \left(t + u, \frac{\rho(t+u)}{\rho(t)}x\right),$$

$$D_u^+ := \{(t, x) \in \Omega_+ : t + u < T_*\}.$$

$$|x| < \rho(t) \implies \left| \frac{\rho(t+u)}{\rho(t)} x \right| < \rho(t+u).$$

$$\Phi_u^- := \tau \circ \Phi_u^+ \circ \tau.$$

$$\Phi_u^-(t, x) = \left(t - u, \frac{\rho(|t-u|)}{\rho(|t|)} x \right), \quad (t < 0).$$

$$\tau \circ \Phi_u^+ = \Phi_u^- \circ \tau.$$

$$\Phi_{u+v}^+ = \Phi_u^+ \circ \Phi_v^+, \quad \Phi_{u+v}^- = \Phi_u^- \circ \Phi_v^-.$$

$$\Phi_u^+(\Phi_v^+(t, x)) = \left(t + u + v, \frac{\rho(t+u+v)}{\rho(t+v)} \frac{\rho(t+v)}{\rho(t)} x \right) = \left(t + u + v, \frac{\rho(t+u+v)}{\rho(t)} x \right) = \Phi_{u+v}^+(t, x).$$

3. Time operators and reflection.

$$(T_u^\pm f)(z) := f(\Phi_u^\pm(z)).$$

$$\mathbb{T}_u^\pm A := (\Phi_u^\pm)^* A.$$

$$\Theta A := \tau^* A.$$

$$(\Theta A)_0(t, x) = -A_0(-t, x), \quad (\Theta A)_j(t, x) = A_j(-t, x), \quad j = 1, 2, 3.$$

$$\Theta \mathbb{T}_u^+ = \mathbb{T}_u^- \Theta.$$

$$\Theta \mathbb{T}_u^+ = \tau^*(\Phi_u^+)^* = (\Phi_u^+ \circ \tau)^* = (\tau \circ \Phi_u^-)^* = (\Phi_u^-)^* \tau^* = \mathbb{T}_u^- \Theta.$$

4. Operator-path measure.

$$G \subset U(N), \quad \mathfrak{g} \subset \mathfrak{u}(N), \quad \mathcal{A}_{\text{hg}}^s := H^s(\Omega_{\text{hg}}; \mathfrak{g} \otimes \mathbb{R}^4), \quad s > 2.$$

$$(\chi_n)_{n \geq 1} \subset L^2(\mathbb{R}^3; \mathfrak{g} \otimes \mathbb{R}^4) \quad \text{orthonormal.}$$

$$c_n(t; A) := \int_{|x| < \rho(|t|)} \langle A(t, x), \chi_n(x) \rangle dx.$$

$$v_A(t) := \sum_{n=1}^{\infty} c_n(t; A) e_n \in K := \ell^2(\mathbb{N}) \quad \text{for a.e. } t.$$

$$\Gamma_A(t)(\lambda u_X) := \lambda v_A(t).$$

$$\Omega_{\text{hg}} = \Omega_- \cup \Sigma_0 \cup \Omega_+, \quad \Omega_- := \Omega_{\text{hg}} \cap \{t < 0\}, \quad \Omega_+ := \Omega_{\text{hg}} \cap \{t > 0\}.$$

$$\theta : \Omega_- \rightarrow \Omega_+, \quad \theta(t, x) = (-t, x).$$

$$\theta^{-1} = \theta.$$

$$K_- := L^2(\Omega_-; \mathfrak{g} \otimes T^*\Omega_-).$$

$$K_+ := L^2(\Omega_+; \mathfrak{g} \otimes T^*\Omega_+).$$

$$\Theta : K_- \rightarrow K_+$$

$$(\Theta A_-)(t, x) := \theta_* A_-(-t, x).$$

$$\Theta \text{ is unitary.}$$

$$Q_- : K_- \rightarrow K_-$$

$$Q_- = Q_-^* \geq 0, \quad \text{Tr}_{K_-} Q_- < \infty.$$

γ_- is the centered Gaussian Borel probability measure on K_- with covariance Q_- .

$$\int_{K_-} e^{i\langle h, A_- \rangle} d\gamma_-(A_-) = \exp\left(-\frac{1}{2}\langle Q_- h, h \rangle\right).$$

$$\gamma_-(K_-) = 1.$$

$$Q_+ := \Theta Q_- \Theta^{-1}.$$

$$\gamma_+ := \Theta_{\#} \gamma_-.$$

γ_+ is the centered Gaussian Borel probability measure on K_+ with covariance Q_+ .

$$\int_{K_+} e^{i\langle h, A_+ \rangle} d\gamma_+(A_+) = \exp\left(-\frac{1}{2}\langle Q_+ h, h \rangle\right).$$

$$K_{\text{sym}} := \{A_- \oplus A_+ \in K_- \oplus K_+ : A_+ = \Theta A_-\}.$$

$$\iota_- : K_- \rightarrow K_{\text{sym}}, \quad \iota_-(A_-) := A_- \oplus \Theta A_-.$$

$$\gamma_{\text{hg}} := (\iota_-)_{\#} \gamma_-.$$

γ_{hg} is a Borel probability measure on K_{sym} .

$$\gamma_{\text{hg}}(K_{\text{sym}}) = 1.$$

$$\Theta_{\text{hg}}(A_- \oplus A_+) := \Theta^{-1}A_+ \oplus \Theta A_-.$$

$$A = A_- \oplus \Theta A_- \implies \Theta_{\text{hg}}A = A.$$

$$(\Theta_{\text{hg}})_{\#}\gamma_{\text{hg}} = \gamma_{\text{hg}}.$$

$$\boxed{\gamma_{\text{hg}} \text{ is reflection invariant.}}$$

$V_- : K_- \rightarrow [0, \infty]$ Borel measurable.

$$V_-(A_-) < \infty \quad \gamma_- \text{-a.e.}$$

$$V_+(A_+) := V_-(\Theta^{-1}A_+).$$

$$A = A_- \oplus \Theta A_-.$$

$$V_{\text{hg}}(A) := V_-(A_-) + V_+(\Theta A_-).$$

$$= V_-(A_-) + V_-(A_-) = 2V_-(A_-).$$

$$0 \leq V_{\text{hg}}(A) < \infty \quad \gamma_{\text{hg}} \text{-a.e.}$$

$$e^{-V_{\text{hg}}(A)} = e^{-2V_-(A_-)}.$$

$$0 < e^{-2V_-(A_-)} \leq 1 \quad \gamma_- \text{-a.e.}$$

$$Z_{\text{hg}} := \int_{K_{\text{sym}}} e^{-V_{\text{hg}}(A)} d\gamma_{\text{hg}}(A).$$

$$= \int_{K_-} e^{-2V_-(A_-)} d\gamma_-(A_-).$$

$$0 \leq e^{-2V_-(A_-)} \leq 1.$$

$$Z_{\text{hg}} \leq \int_{K_-} 1 d\gamma_-(A_-) = 1.$$

$$V_-(A_-) < \infty \quad \gamma_- \text{-a.e.}$$

$$\implies e^{-2V_-(A_-)} > 0 \quad \gamma_- \text{-a.e.}$$

$$\gamma_-(K_-) = 1.$$

$$\implies Z_{\text{hg}} > 0.$$

$$\boxed{0 < Z_{\text{hg}} \leq 1 < \infty.}$$

$$\boxed{e^{-V_{\text{hg}}} \in L^1(K_{\text{sym}}, \gamma_{\text{hg}}).}$$

$$d\mu_{\text{hg}}(A) := Z_{\text{hg}}^{-1} e^{-V_{\text{hg}}(A)} d\gamma_{\text{hg}}(A).$$

$$\mu_{\text{hg}}(K_{\text{sym}}) = Z_{\text{hg}}^{-1} \int_{K_{\text{sym}}} e^{-V_{\text{hg}}} d\gamma_{\text{hg}} = 1.$$

$$\boxed{\mu_{\text{hg}} \text{ is a Borel probability measure on } K_{\text{sym}}.}$$

$$(\Theta_{\text{hg}})_{\#} \gamma_{\text{hg}} = \gamma_{\text{hg}}.$$

$$V_{\text{hg}}(\Theta_{\text{hg}} A) = V_{\text{hg}}(A).$$

$$\implies (\Theta_{\text{hg}})_{\#} \mu_{\text{hg}} = \mu_{\text{hg}}.$$

$$\boxed{\mu_{\text{hg}} \text{ is reflection invariant.}}$$

$$F \in L^2(K_{\text{sym}}, \mu_{\text{hg}}), \quad F = F(A_+) \quad \text{depends only on the positive-time component.}$$

$$A = A_- \oplus \Theta A_-.$$

$$(\Theta_{\text{hg}} F)(A) := F(\Theta_{\text{hg}} A).$$

$$\Theta_{\text{hg}} A = A.$$

$$\int \overline{F(A)} (\Theta_{\text{hg}} F)(A) d\mu_{\text{hg}}(A) = \int |F(A)|^2 d\mu_{\text{hg}}(A).$$

$$\geq 0.$$

$$\boxed{\mu_{\text{hg}} \text{ is reflection positive on the symmetric hourglass sector.}}$$

$$\Omega_- := \Omega_{\text{hg}} \cap \{t \leq 0\}.$$

$$\mathcal{H}_- := L^2(\Omega_-; \mathfrak{g} \otimes T^* \Omega_-).$$

$$\mathcal{A}_- := H^1(\Omega_-; \mathfrak{g} \otimes T^*\Omega_-).$$

$$B_- := (I + \nabla_-^* \nabla_-)^{1/2}.$$

$$\mathcal{A}_- = \text{Dom}(B_-).$$

$$\|A_-\|_{\mathcal{A}_-}^2 = \|B_- A_-\|_{\mathcal{H}_-}^2.$$

$$L_- : \text{Dom}(L_-) \subset \mathcal{H}_- \rightarrow \mathcal{H}_-$$

$$L_- = L_-^*, \quad L_- \geq mI \quad (m > 0).$$

$$T_-(\tau) := e^{-\tau L_-}, \quad \tau \geq 0.$$

$$\|T_-(\tau)\|_{\mathcal{H}_- \rightarrow \mathcal{H}_-} \leq e^{-m\tau}.$$

$$C_- : \mathcal{H}_- \rightarrow \mathcal{H}_- \quad \text{Hilbert-Schmidt.}$$

$$Q_- := \int_0^\infty T_-(\tau) C_- C_-^* T_-(\tau)^* d\tau.$$

$$Q_- \geq 0.$$

$$Q_-^* = Q_-.$$

$$\text{Tr}_{\mathcal{H}_-} Q_- = \int_0^\infty \text{Tr} (T_-(\tau) C_- C_-^* T_-(\tau)^*) d\tau.$$

$$= \int_0^\infty \|T_-(\tau) C_-\|_{\text{HS}}^2 d\tau.$$

$$\|T_-(\tau) C_-\|_{\text{HS}}^2 \leq \|T_-(\tau)\|^2 \|C_-\|_{\text{HS}}^2.$$

$$\leq e^{-2m\tau} \|C_-\|_{\text{HS}}^2.$$

$$\text{Tr} Q_- \leq \int_0^\infty e^{-2m\tau} \|C_-\|_{\text{HS}}^2 d\tau.$$

$$= \frac{1}{2m} \|C_-\|_{\text{HS}}^2 < \infty.$$

$$Q_- \quad \text{is positive trace class.}$$

γ_- is the centered Gaussian Borel probability measure on $\mathcal{H}_-$ with covariance Q_- .

$$\int_{\mathcal{H}_-} e^{i\langle h, A_- \rangle_{\mathcal{H}_-}} d\gamma_-(A_-) = \exp\left(-\frac{1}{2}\langle Q_- h, h \rangle_{\mathcal{H}_-}\right).$$

$$\gamma_-(\mathcal{H}_-) = 1.$$

$$\mathrm{Tr}_{\mathcal{H}_-}(B_- Q_- B_-) < \infty.$$

$$\int_{\mathcal{H}_-} \|B_- A_-\|_{\mathcal{H}_-}^2 d\gamma_-(A_-) = \mathrm{Tr}(B_- Q_- B_-).$$

$$< \infty.$$

$$\implies \|B_- A_-\|_{\mathcal{H}_-} < \infty \quad \gamma_- \text{-a.e.}$$

$$\implies A_- \in \mathrm{Dom}(B_-) = \mathcal{A}_- \quad \gamma_- \text{-a.e.}$$

$$\boxed{\gamma_-(\mathcal{A}_-) = 1.}$$

$$A_- \in H^1(\Omega_-) \implies dA_- \in L^2(\Omega_-).$$

$$\Omega_- \subset \mathbb{R}^4.$$

$$H^1(\Omega_-) \hookrightarrow L^4(\Omega_-).$$

$$A_- \in H^1(\Omega_-) \implies A_- \in L^4(\Omega_-).$$

$$|A_- \wedge A_-| \leq C_{\mathfrak{g}} |A_-|^2.$$

$$\|A_- \wedge A_-\|_{L^2(\Omega_-)}^2 \leq C_{\mathfrak{g}}^2 \int_{\Omega_-} |A_-|^4 dx.$$

$$= C_{\mathfrak{g}}^2 \|A_-\|_{L^4(\Omega_-)}^4.$$

$$< \infty.$$

$$F_{A_-} := dA_- + A_- \wedge A_-.$$

$$dA_- \in L^2(\Omega_-), \quad A_- \wedge A_- \in L^2(\Omega_-).$$

$$\implies F_{A_-} \in L^2(\Omega_-).$$

$$V_-(A_-) := \frac{1}{2g^2} \int_{\Omega_-} |F_{A_-}(x)|^2 dx.$$

$$F_{A_-} \in L^2(\Omega_-) \implies 0 \leq V_-(A_-) < \infty.$$

$$\gamma_-(\mathcal{A}_-) = 1.$$

$$\implies 0 \leq V_-(A_-) < \infty \quad \gamma_- \text{-a.e.}$$

$$\boxed{V_-(A_-) < \infty \quad \gamma_- \text{-a.e.}}$$

$$0 \leq V_-(A_-) < \infty \quad \gamma_- \text{-a.e.}$$

$$\implies 0 < e^{-V_-(A_-)} \leq 1 \quad \gamma_- \text{-a.e.}$$

$$Z_- := \int_{\mathcal{H}_-} e^{-V_-(A_-)} d\gamma_-(A_-).$$

$$Z_- \leq \int_{\mathcal{H}_-} 1 d\gamma_-(A_-) = 1.$$

$$e^{-V_-(A_-)} > 0 \quad \gamma_- \text{-a.e.}$$

$$\gamma_-(\mathcal{H}_-) = 1.$$

$$\implies Z_- > 0.$$

$$\boxed{0 < Z_- \leq 1 < \infty.}$$

$$\boxed{e^{-V_-} \in L^1(\gamma_-).}$$

$$d\mu_-(A_-) := Z_-^{-1} e^{-V_-(A_-)} d\gamma_-(A_-).$$

$$\mu_-(\mathcal{H}_-) = Z_-^{-1} \int_{\mathcal{H}_-} e^{-V_-(A_-)} d\gamma_-(A_-) = 1.$$

$$\boxed{\mu_- \text{ is a Borel probability measure on } \mathcal{H}_-.}$$

$$\Omega_- := \Omega_{\text{hg}} \cap \{t \leq 0\}.$$

$$\mathcal{H}_- := L^2(\Omega_-; \mathfrak{g} \otimes T^*\Omega_-).$$

$$\mathfrak{l}_-(u, u) := \int_{\Omega_-} (|\nabla_{\text{hg}} u|^2 + m^2 |u|^2) d\text{vol}_{\text{hg}}, \quad m > 0.$$

$$\text{Dom}(\mathfrak{l}_-) = H^1(\Omega_-; \mathfrak{g} \otimes T^*\Omega_-).$$

$$\mathfrak{l}_-(u, u) \geq m^2 \|u\|_{\mathcal{H}_-}^2.$$

$\mathfrak{l}_-$ is densely defined, closed, symmetric, and lower semibounded.

$$\implies \exists! L_- = L_-^* \quad \text{associated with } \mathfrak{L}_-.$$

$$L_- \geq m^2 I.$$

$$T_-(\tau) := e^{-\tau L_-}, \quad \tau \geq 0.$$

$$\|T_-(\tau)\|_{\mathcal{H}_- \rightarrow \mathcal{H}_-} \leq e^{-m^2 \tau}.$$

Ω_- is hourglass-compactified with bounded Lipschitz slices.

$$\implies (L_- + I)^{-1} \text{ is compact.}$$

$$\implies \exists \{e_n\}_{n=1}^\infty \subset \text{Dom}(L_-), \quad L_- e_n = \lambda_n e_n,$$

$$0 < m^2 \leq \lambda_1 \leq \lambda_2 \leq \cdots, \quad \lambda_n \rightarrow \infty,$$

$$\{e_n\}_{n=1}^\infty \text{ is an orthonormal basis of } \mathcal{H}_-.$$

$$\dim \Omega_- = 4.$$

$$\lambda_n \sim c n^{1/2} \quad (n \rightarrow \infty).$$

$$\beta > \frac{3}{2}.$$

$$C_- := L_-^{-\beta}.$$

$$C_- e_n = \lambda_n^{-\beta} e_n.$$

$$C_- C_-^* = L_-^{-2\beta}.$$

$$Q_- := \int_0^\infty T_-(\tau) C_- C_-^* T_-(\tau)^* d\tau.$$

$$= \int_0^\infty e^{-\tau L_-} L_-^{-2\beta} e^{-\tau L_-} d\tau.$$

$$= \int_0^\infty e^{-2\tau L_-} L_-^{-2\beta} d\tau.$$

$$Q_- e_n = \left(\int_0^\infty e^{-2\tau \lambda_n} \lambda_n^{-2\beta} d\tau \right) e_n.$$

$$= \frac{1}{2} \lambda_n^{-(2\beta+1)} e_n.$$

$$Q_- = \frac{1}{2} L_-^{-(2\beta+1)}.$$

$$Q_- \geq 0, \quad Q_-^* = Q_-.$$

$$\mathrm{Tr} Q_- = \frac{1}{2} \sum_{n=1}^{\infty} \lambda_n^{-(2\beta+1)}.$$

$$\lambda_n \sim c n^{1/2} \implies \lambda_n^{-(2\beta+1)} \sim C n^{-(2\beta+1)/2}.$$

$$\beta > \frac{1}{2} \implies \frac{2\beta+1}{2} > 1.$$

$$\implies \sum_{n=1}^{\infty} \lambda_n^{-(2\beta+1)} < \infty.$$

$$\implies \mathrm{Tr} Q_- < \infty.$$

Q_- is positive trace class.

$$B_- := (I + L_-)^{1/2}.$$

$$\mathrm{Dom}(B_-) = H^1(\Omega_-; \mathfrak{g} \otimes T^* \Omega_-).$$

$$B_- Q_- B_- = \frac{1}{2} (I + L_-)^{1/2} L_-^{-(2\beta+1)} (I + L_-)^{1/2}.$$

$$B_- Q_- B_- = \frac{1}{2} (I + L_-) L_-^{-(2\beta+1)}.$$

$$\mathrm{Tr}(B_- Q_- B_-) = \frac{1}{2} \sum_{n=1}^{\infty} (1 + \lambda_n) \lambda_n^{-(2\beta+1)}.$$

$$= \frac{1}{2} \sum_{n=1}^{\infty} \left(\lambda_n^{-(2\beta+1)} + \lambda_n^{-2\beta} \right).$$

$$\lambda_n^{-2\beta} \sim C n^{-\beta}.$$

$$\beta > \frac{3}{2} \implies \beta > 1.$$

$$\implies \sum_{n=1}^{\infty} \lambda_n^{-2\beta} < \infty.$$

$$\sum_{n=1}^{\infty} \lambda_n^{-(2\beta+1)} < \infty.$$

$$\implies \mathrm{Tr}(B_- Q_- B_-) < \infty.$$

$$\boxed{\mathrm{Tr}(B_- Q_- B_-) < \infty.}$$

$$\|B_- T_-(\tau) C_-\|_{\mathrm{HS}}^2 = \sum_{n=1}^{\infty} \|B_- e^{-\tau L_-} L_-^{-\beta} e_n\|^2.$$

$$= \sum_{n=1}^{\infty} (1 + \lambda_n) e^{-2\tau \lambda_n} \lambda_n^{-2\beta}.$$

$$\leq e^{-2m^2 \tau} \sum_{n=1}^{\infty} (1 + \lambda_n) \lambda_n^{-2\beta}.$$

$$\sum_{n=1}^{\infty} (1 + \lambda_n) \lambda_n^{-2\beta} = \sum_{n=1}^{\infty} (\lambda_n^{-2\beta} + \lambda_n^{1-2\beta}).$$

$$\lambda_n^{1-2\beta} \sim C n^{(1-2\beta)/2}.$$

$$\beta > \frac{3}{2} \implies \frac{1-2\beta}{2} < -1.$$

$$\implies \sum_{n=1}^{\infty} \lambda_n^{1-2\beta} < \infty.$$

$$\sum_{n=1}^{\infty} \lambda_n^{-2\beta} < \infty.$$

$$M := \sum_{n=1}^{\infty} (1 + \lambda_n) \lambda_n^{-2\beta} < \infty.$$

$$\boxed{\|B_- T_-(\tau) C_-\|_{\mathrm{HS}}^2 \leq M e^{-2m^2 \tau}.$$

$$\alpha := 0.$$

$$0 \leq \alpha < 1.$$

$$M e^{-2m^2 \tau} \leq M(1 + \tau^{-\alpha}) e^{-2m^2 \tau}.$$

$$\boxed{\|B_- T_-(\tau) C_-\|_{\mathrm{HS}}^2 \leq M(1 + \tau^{-\alpha}) e^{-2m^2 \tau}, \quad 0 \leq \alpha < 1.}$$

$$\mathrm{Tr}(B_- Q_- B_-) = \int_0^{\infty} \|B_- T_-(\tau) C_-\|_{\mathrm{HS}}^2 d\tau.$$

$$\leq M \int_0^{\infty} e^{-2m^2 \tau} d\tau.$$

$$= \frac{M}{2m^2} < \infty.$$

$$\boxed{\mathrm{Tr}(B_- Q_- B_-) < \infty.}$$

$$\gamma_- := \mathcal{N}(0, Q_-).$$

$$\int_{\mathcal{H}_-} \|B_- A_-\|_{\mathcal{H}_-}^2 d\gamma_-(A_-) = \text{Tr}(B_- Q_- B_-).$$

$$< \infty.$$

$$\implies \|B_- A_-\|_{\mathcal{H}_-} < \infty \quad \gamma_- \text{-a.e.}$$

$$\implies A_- \in \text{Dom}(B_-) = H^1(\Omega_-; \mathfrak{g} \otimes T^* \Omega_-) \quad \gamma_- \text{-a.e.}$$

$$\boxed{\gamma_-(H^1(\Omega_-; \mathfrak{g} \otimes T^* \Omega_-)) = 1.}$$

$$A_- \in H^1(\Omega_-) \implies dA_- \in L^2(\Omega_-).$$

$$H^1(\Omega_-) \hookrightarrow L^4(\Omega_-).$$

$$A_- \in H^1(\Omega_-) \implies A_- \in L^4(\Omega_-).$$

$$|A_- \wedge A_-| \leq C_{\mathfrak{g}} |A_-|^2.$$

$$\|A_- \wedge A_-\|_{L^2(\Omega_-)}^2 \leq C_{\mathfrak{g}}^2 \|A_-\|_{L^4(\Omega_-)}^4.$$

$$< \infty.$$

$$F_{A_-} := dA_- + A_- \wedge A_-.$$

$$dA_- \in L^2(\Omega_-), \quad A_- \wedge A_- \in L^2(\Omega_-).$$

$$\implies F_{A_-} \in L^2(\Omega_-).$$

$$V_-(A_-) := \frac{1}{2g^2} \int_{\Omega_-} |F_{A_-}(x)|^2 dx.$$

$$0 \leq V_-(A_-) < \infty \quad \gamma_- \text{-a.e.}$$

$$0 < e^{-V_-(A_-)} \leq 1 \quad \gamma_- \text{-a.e.}$$

$$Z_- := \int_{\mathcal{H}_-} e^{-V_-(A_-)} d\gamma_-(A_-).$$

$$0 < Z_- \leq 1 < \infty.$$

$$\boxed{e^{-V_-} \in L^1(\gamma_-), \quad 0 < Z_- < \infty.}$$

The smoothing trace bound is not assumed; it is generated by the inverse-time hourglass operator L_- .

$$\Omega_- := \Omega_{\text{hg}} \cap \{t \leq 0\}.$$

$$\mathcal{H}_- := L^2(\Omega_-; \mathfrak{g} \otimes T^*\Omega_-).$$

$$\mathfrak{l}_-(u, u) := \int_{\Omega_-} (|\nabla_{\text{hg}} u|^2 + m^2 |u|^2) d\text{vol}_{\text{hg}}, \quad m > 0.$$

$$\text{Dom}(\mathfrak{l}_-) = H^1(\Omega_-; \mathfrak{g} \otimes T^*\Omega_-).$$

$$\mathfrak{l}_-(u, u) \geq m^2 \|u\|_{\mathcal{H}_-}^2.$$

$\mathfrak{l}_-$ is densely defined, closed, symmetric, lower semibounded.

$$\implies \exists! L_- = L_-^* \quad \text{associated with } \mathfrak{l}_-.$$

$$L_- \geq m^2 I.$$

$$T_-(\tau) := e^{-\tau L_-}, \quad \tau \geq 0.$$

$$\|T_-(\tau)\| \leq e^{-m^2 \tau}.$$

$$(L_- + I)^{-1} \text{ is compact.}$$

$$\implies \exists \{e_n\}_{n=1}^\infty, \quad L_- e_n = \lambda_n e_n,$$

$$0 < m^2 \leq \lambda_1 \leq \lambda_2 \leq \dots, \quad \lambda_n \rightarrow \infty,$$

$$\{e_n\}_{n=1}^\infty \text{ is an orthonormal basis of } \mathcal{H}_-.$$

$$\dim \Omega_- = 4.$$

$$\lambda_n \sim c n^{1/2}.$$

$$C_{\varepsilon, -} := \sqrt{2} e^{-\frac{\varepsilon}{2} L_-}, \quad \varepsilon > 0.$$

$$C_{\varepsilon, -} C_{\varepsilon, -}^* = 2e^{-\varepsilon L_-}.$$

$$Q_{\varepsilon, -} := \int_0^\infty T_-(\tau) C_{\varepsilon, -} C_{\varepsilon, -}^* T_-(\tau)^* d\tau.$$

$$\begin{aligned}
&= \int_0^\infty e^{-\tau L_-} 2e^{-\varepsilon L_-} e^{-\tau L_-} d\tau. \\
&= 2 \int_0^\infty e^{-(2\tau+\varepsilon)L_-} d\tau. \\
Q_{\varepsilon,-} e_n &= 2 \left(\int_0^\infty e^{-(2\tau+\varepsilon)\lambda_n} d\tau \right) e_n. \\
&= 2e^{-\varepsilon\lambda_n} \left(\int_0^\infty e^{-2\tau\lambda_n} d\tau \right) e_n. \\
&= 2e^{-\varepsilon\lambda_n} \frac{1}{2\lambda_n} e_n. \\
&= e^{-\varepsilon\lambda_n} \lambda_n^{-1} e_n.
\end{aligned}$$

$$\boxed{Q_{\varepsilon,-} = e^{-\varepsilon L_-} L_-^{-1}.}$$

$$G_- := L_-^{-1}.$$

$$\boxed{Q_{\varepsilon,-} = e^{-\varepsilon L_-} G_-}.$$

Thus $Q_{\varepsilon,-}$ is the heat-regularized inverse-time propagator.

$$Q_{\varepsilon,-} \geq 0, \quad Q_{\varepsilon,-} = Q_{\varepsilon,-}^*.$$

$$\mathrm{Tr} Q_{\varepsilon,-} = \sum_{n=1}^\infty e^{-\varepsilon\lambda_n} \lambda_n^{-1}.$$

$$\lambda_n \sim c n^{1/2}.$$

$$e^{-\varepsilon\lambda_n} \lambda_n^{-1} \leq C_\varepsilon e^{-c_\varepsilon n^{1/2}} n^{-1/2}.$$

$$\implies \sum_{n=1}^\infty e^{-\varepsilon\lambda_n} \lambda_n^{-1} < \infty.$$

$$\boxed{Q_{\varepsilon,-} \text{ is positive trace class for every } \varepsilon > 0.}$$

$$B_- := (I + L_-)^{1/2}.$$

$$\mathrm{Dom}(B_-) = H^1(\Omega_-; \mathfrak{g} \otimes T^* \Omega_-).$$

$$B_- Q_{\varepsilon,-} B_- = (I + L_-)^{1/2} e^{-\varepsilon L_-} L_-^{-1} (I + L_-)^{1/2}.$$

$$= (I + L_-) e^{-\varepsilon L_-} L_-^{-1}.$$

$$\mathrm{Tr}(B_- Q_{\varepsilon, -} B_-) = \sum_{n=1}^{\infty} (1 + \lambda_n) e^{-\varepsilon \lambda_n} \lambda_n^{-1}.$$

$$= \sum_{n=1}^{\infty} e^{-\varepsilon \lambda_n} + \sum_{n=1}^{\infty} e^{-\varepsilon \lambda_n} \lambda_n^{-1}.$$

$$\lambda_n \sim c n^{1/2} \implies \sum_{n=1}^{\infty} e^{-\varepsilon \lambda_n} < \infty.$$

$$\sum_{n=1}^{\infty} e^{-\varepsilon \lambda_n} \lambda_n^{-1} < \infty.$$

$$\boxed{\mathrm{Tr}(B_- Q_{\varepsilon, -} B_-) < \infty.}$$

$$\gamma_{\varepsilon, -} := \mathcal{N}(0, Q_{\varepsilon, -}).$$

$$\int_{\mathcal{H}_-} \|B_- A_-\|_{\mathcal{H}_-}^2 d\gamma_{\varepsilon, -}(A_-) = \mathrm{Tr}(B_- Q_{\varepsilon, -} B_-) < \infty.$$

$$\implies A_- \in H^1(\Omega_-; \mathfrak{g} \otimes T^* \Omega_-) \quad \gamma_{\varepsilon, -}\text{-a.e.}$$

$$A_- \in H^1(\Omega_-) \implies dA_- \in L^2(\Omega_-).$$

$$H^1(\Omega_-) \hookrightarrow L^4(\Omega_-).$$

$$A_- \in H^1(\Omega_-) \implies A_- \in L^4(\Omega_-).$$

$$|A_- \wedge A_-| \leq C_{\mathfrak{g}} |A_-|^2.$$

$$\|A_- \wedge A_-\|_{L^2(\Omega_-)}^2 \leq C_{\mathfrak{g}}^2 \|A_-\|_{L^4(\Omega_-)}^4 < \infty.$$

$$F_{A_-} := dA_- + A_- \wedge A_-.$$

$$F_{A_-} \in L^2(\Omega_-) \quad \gamma_{\varepsilon, -}\text{-a.e.}$$

$$V_{\varepsilon, -}(A_-) := \frac{1}{2g^2} \int_{\Omega_-} |F_{A_-}(x)|^2 dx.$$

$$0 \leq V_{\varepsilon, -}(A_-) < \infty \quad \gamma_{\varepsilon, -}\text{-a.e.}$$

$$Z_{\varepsilon, -} := \int_{\mathcal{H}_-} e^{-V_{\varepsilon, -}(A_-)} d\gamma_{\varepsilon, -}(A_-).$$

$$0 < e^{-V_{\varepsilon, -}(A_-)} \leq 1 \quad \gamma_{\varepsilon, -}\text{-a.e.}$$

$$0 < Z_{\varepsilon, -} \leq 1 < \infty.$$

$$\boxed{e^{-V_{\varepsilon,-}} \in L^1(\gamma_{\varepsilon,-}), \quad 0 < Z_{\varepsilon,-} < \infty.}$$

$$d\mu_{\varepsilon,-}(A_-) := Z_{\varepsilon,-}^{-1} e^{-V_{\varepsilon,-}(A_-)} d\gamma_{\varepsilon,-}(A_-).$$

$$\boxed{\mu_{\varepsilon,-} \text{ is a Borel probability measure.}}$$

$$\Omega_{\text{hg}} = \Omega_- \cup \Sigma_0 \cup \Omega_+.$$

$$\theta : \Omega_- \rightarrow \Omega_+, \quad \theta(t, x) = (-t, x).$$

$$\mu_{\varepsilon,-} = Z_{\varepsilon,-}^{-1} e^{-V_{\varepsilon,-}} d\gamma_{\varepsilon,-}.$$

$$\mu_{\varepsilon,+} := \theta_{\#} \mu_{\varepsilon,-}.$$

$$V_{\varepsilon,+}(A_+) := V_{\varepsilon,-}(\theta^{-1}A_+).$$

$$\gamma_{\varepsilon,+} := \theta_{\#} \gamma_{\varepsilon,-}.$$

$$d\mu_{\varepsilon,+}(A_+) = Z_{\varepsilon,-}^{-1} e^{-V_{\varepsilon,+}(A_+)} d\gamma_{\varepsilon,+}(A_+).$$

$$Z_{\varepsilon,+} = \int e^{-V_{\varepsilon,+}} d\gamma_{\varepsilon,+} = \int e^{-V_{\varepsilon,-}} d\gamma_{\varepsilon,-} = Z_{\varepsilon,-}.$$

$$\boxed{\mu_{\varepsilon,+} = \theta_{\#} \mu_{\varepsilon,-}.$$

$$0 < Z_{\varepsilon,-} < \infty \implies 0 < Z_{\varepsilon,+} < \infty.$$

$$e^{-V_{\varepsilon,-}} \in L^1(\gamma_{\varepsilon,-}) \implies e^{-V_{\varepsilon,+}} \in L^1(\gamma_{\varepsilon,+}).$$

$$Q_{\varepsilon,-} = e^{-\varepsilon L_-} L_-^{-1}.$$

$$G_- := L_-^{-1}.$$

$$Q_{\varepsilon,-} - G_- = (e^{-\varepsilon L_-} - I) L_-^{-1}.$$

$$\|(Q_{\varepsilon,-} - G_-)\| = \sup_{\lambda \in \sigma(L_-)} \left| (e^{-\varepsilon \lambda} - 1) \lambda^{-1} \right|.$$

$$1 - e^{-\varepsilon \lambda} \leq \varepsilon \lambda.$$

$$\left| (e^{-\varepsilon \lambda} - 1) \lambda^{-1} \right| \leq \varepsilon.$$

$$\implies \|Q_{\varepsilon,-} - G_-\| \leq \varepsilon.$$

$$\boxed{Q_{\varepsilon,-} \rightarrow G_- = L_-^{-1} \quad \text{in operator norm on } \mathcal{H}_-.}$$

$$f, g \in \mathcal{H}_-.$$

$$|\langle f, Q_{\varepsilon,-}g \rangle - \langle f, G_-g \rangle| \leq \|f\| \|Q_{\varepsilon,-} - G_-\| \|g\|.$$

$$\leq \varepsilon \|f\| \|g\|.$$

$$\implies \quad \langle f, Q_{\varepsilon,-}g \rangle \rightarrow \langle f, G_-g \rangle.$$

$$\boxed{\langle f, Q_{\varepsilon,-}g \rangle \rightarrow \langle f, L_-^{-1}g \rangle.}$$

$$\int e^{i\langle f, A_- \rangle} d\gamma_{\varepsilon,-}(A_-) = \exp\left(-\frac{1}{2}\langle f, Q_{\varepsilon,-}f \rangle\right).$$

$$\langle f, Q_{\varepsilon,-}f \rangle \rightarrow \langle f, L_-^{-1}f \rangle.$$

$$\implies \quad \int e^{i\langle f, A_- \rangle} d\gamma_{\varepsilon,-}(A_-) \rightarrow \exp\left(-\frac{1}{2}\langle f, L_-^{-1}f \rangle\right).$$

$$\boxed{\gamma_{\varepsilon,-} \implies \gamma_{0,-} \text{ in all finite-dimensional cylinder distributions,}}$$

$$\boxed{\text{Cov}(\gamma_{0,-}) = L_-^{-1}.}$$

$$\boxed{L_-^{-1} \text{ is the standard gauge-fixed free Yang-Mills propagator on } \Omega_- .}$$

$$s > 1.$$

$$\mathcal{H}_-^{-s} := \text{Dom}((I + L_-)^{-s/2})^*.$$

$$\text{Tr}_{\mathcal{H}_-^{-s}}(L_-^{-1}) = \sum_{n=1}^{\infty} (1 + \lambda_n)^{-s} \lambda_n^{-1}.$$

$$\lambda_n \sim c n^{1/2}.$$

$$(1 + \lambda_n)^{-s} \lambda_n^{-1} \sim C n^{-(s+1)/2}.$$

$$s > 1 \quad \implies \quad \frac{s+1}{2} > 1.$$

$$\implies \text{Tr}_{\mathcal{H}_-^{-s}}(L_-^{-1}) < \infty.$$

$$\implies \exists \gamma_{0,-} = \mathcal{N}(0, L_-^{-1}) \quad \text{as a Borel Gaussian measure on } \mathcal{H}_-^{-s}.$$

$$\gamma_{\varepsilon,-} \implies \gamma_{0,-} \quad \text{weakly on } \mathcal{H}_-^{-s} \quad (s > 1).$$

$Q_{\varepsilon,-}$ is a regularized L^2 -covariance, $Q_{\varepsilon,-} \rightarrow L_-^{-1}$ as the standard distributional YM propagator.

$$\text{At the Gaussian reference level,} \quad Q_{\varepsilon,-} \rightarrow L_-^{-1}.$$

Thus the two-point function converges to the standard gauge-fixed free YM propagator.

The interacting hourglass YM measure is obtained by the Gibbs density $e^{-V_{\varepsilon,-}}$.

$$\mu_{\varepsilon,-} \rightsquigarrow \mathcal{Q}_{\varepsilon}^{R,\rho} \xrightarrow{\text{Mosco}} \mathcal{Q}_0^{R,\rho}.$$

$$\nu_{\text{op}} := (\Gamma_{\bullet})_{\#} \mu.$$

$$\Gamma_T : \mathcal{A} \rightarrow C([-T, T]; \mathcal{A}), \quad \Gamma_T(A)(t) := U_t A.$$

$$\nu_T := (\Gamma_T)_{\#} \mu.$$

$$\text{ev}_t : C([-T, T]; \mathcal{A}) \rightarrow \mathcal{A}, \quad \text{ev}_t(\gamma) = \gamma(t).$$

$$\text{ev}_t \circ \Gamma_T = U_t.$$

$$\nu_T(B) = \mu(\Gamma_T^{-1}(B)).$$

$$\Theta_{\#} \mu = \mu, \quad (\Re \gamma)(t) := \Theta \gamma(-t).$$

$$\Re(\Gamma_T(A))(t) = \Theta(U_{-t}A) = \Theta(\Theta U_t \Theta A) = U_t(\Theta A) = \Gamma_T(\Theta A)(t).$$

$$\Re \circ \Gamma_T = \Gamma_T \circ \Theta.$$

$$\Re_{\#} \nu_T = \Re_{\#} (\Gamma_T)_{\#} \mu = (\Gamma_T)_{\#} \Theta_{\#} \mu = (\Gamma_T)_{\#} \mu = \nu_T.$$

5. Wilson holonomy reflection identity.

$$A = \sum_{\mu=0}^3 A_{\mu} dx^{\mu}, \quad \gamma : [0, 1] \rightarrow \Omega_{\text{hg}} \quad \text{piecewise absolutely continuous.}$$

$$B_{A,\gamma}(s) := \sum_{\mu=0}^3 A_{\mu}(\gamma(s)) \dot{\gamma}^{\mu}(s).$$

$$B_{A,\gamma}(s) \in \mathfrak{u}(N) \quad \text{for a.e. } s.$$

$$\frac{d}{ds} U_{A,\gamma}(s) = -B_{A,\gamma}(s) U_{A,\gamma}(s), \quad U_{A,\gamma}(0) = I_N.$$

$$\gamma^\Theta(s) := \tau(\gamma(1-s)).$$

$$\dot{\gamma}^{\Theta,0}(s) = \dot{\gamma}^0(1-s), \quad \dot{\gamma}^{\Theta,j}(s) = -\dot{\gamma}^j(1-s), \quad j = 1, 2, 3.$$

$$\begin{aligned} B_{\Theta A, \gamma^\Theta}(s) &= (\Theta A)_0(\gamma^\Theta(s)) \dot{\gamma}^{\Theta,0}(s) + \sum_{j=1}^3 (\Theta A)_j(\gamma^\Theta(s)) \dot{\gamma}^{\Theta,j}(s) \\ &= (-A_0(\gamma(1-s))) \dot{\gamma}^0(1-s) + \sum_{j=1}^3 A_j(\gamma(1-s)) (-\dot{\gamma}^j(1-s)) \\ &= -\sum_{\mu=0}^3 A_\mu(\gamma(1-s)) \dot{\gamma}^\mu(1-s) = -B_{A,\gamma}(1-s). \end{aligned}$$

$$V(s) := U_{A,\gamma}(1-s) U_{A,\gamma}(1)^{-1}.$$

$$\frac{d}{ds} V(s) = B_{A,\gamma}(1-s) V(s) = -B_{\Theta A, \gamma^\Theta}(s) V(s), \quad V(0) = I_N.$$

$$U_{\Theta A, \gamma^\Theta}(s) = V(s).$$

$$U_{\Theta A, \gamma^\Theta}(1) = U_{A,\gamma}(1)^{-1} = U_{A,\gamma}(1)^*.$$

$$W_\gamma(A) := \frac{1}{N} \text{Tr}(U_{A,\gamma}(1)).$$

$$W_{\gamma^\Theta}(\Theta A) = \frac{1}{N} \text{Tr}(U_{\Theta A, \gamma^\Theta}(1)) = \frac{1}{N} \text{Tr}(U_{A,\gamma}(1)^*) = \overline{W_\gamma(A)}.$$

$$U_{\Theta A, \gamma^\Theta} = U_{A,\gamma}^{-1} = U_{A,\gamma}^*, \quad W_{\gamma^\Theta}(\Theta A) = \overline{W_\gamma(A)}.$$

6. Gaussian reflection positivity.

$$\mathcal{H} := L^2(\Omega_{\text{hg}}; \mathfrak{g} \otimes \mathbb{R}^4), \quad \mathcal{H}_+ := \{f \in \mathcal{H} : \text{supp}(f) \subset \overline{\Omega_+}\}.$$

$$\gamma_Q = N(0, Q), \quad Q\Theta = \Theta Q, \quad \langle f, Q\Theta f \rangle_{\mathcal{H}} \geq 0 \quad (\forall f \in \mathcal{H}_+).$$

$$\mathcal{E}_f(A) := \exp\left(i\langle A, f \rangle + \frac{1}{2}\langle f, Qf \rangle\right).$$

$$\int \overline{\mathcal{E}_{\Theta f}(A)} \mathcal{E}_g(A) d\gamma_Q(A) = \exp(\langle f, Q\Theta g \rangle).$$

$$M_{ij} := \langle f_i, Q\Theta f_j \rangle, \quad f_1, \dots, f_m \in \mathcal{H}_+.$$

$$\sum_{i,j=1}^m \bar{c}_i c_j M_{ij} = \left\langle \sum_i c_i f_i, Q\Theta \left(\sum_j c_j f_j \right) \right\rangle \geq 0.$$

$$(M_{ij})_{i,j=1}^m \geq 0.$$

$$(e^{M_{ij}})_{i,j=1}^m \geq 0.$$

$$F = \sum_{j=1}^m c_j \mathcal{E}_{f_j}.$$

$$\int \overline{F(\Theta A)} F(A) d\gamma_Q(A) = \sum_{i,j=1}^m \bar{c}_i c_j e^{M_{ij}} \geq 0.$$

7. OS reconstruction of the transfer Hamiltonian.

$\mathfrak{A}_+$ positive-time observable algebra.

$$\langle F, G \rangle_{\text{OS}} := \int \overline{F(\Theta A)} G(A) d\mu(A).$$

$$\langle F, F \rangle_{\text{OS}} \geq 0 \quad (\forall F \in \mathfrak{A}_+).$$

$$\mathcal{N} := \{F \in \mathfrak{A}_+ : \langle F, F \rangle_{\text{OS}} = 0\}.$$

$$\mathcal{H}_{\text{phys}} := \overline{\mathfrak{A}_+ / \mathcal{N}}^{\|\cdot\|_{\text{OS}}}.$$

$$(S_t F)(A) := F(U_t A), \quad t \geq 0.$$

$$\widehat{S}_t[F] := [S_t F].$$

$$\Theta U_t \Theta = U_{-t}.$$

$$\langle \widehat{S}_t[F], [G] \rangle_{\text{phys}} = \langle [F], \widehat{S}_t[G] \rangle_{\text{phys}}.$$

$$\|\widehat{S}_t\| \leq 1.$$

$$\widehat{S}_t = e^{-tH}, \quad H = H^*, \quad H \geq 0.$$

8. Positive-time measure and electric transfer form.

$\mathfrak{A}_+ :=$ positive-time observable algebra.

$$\langle F, G \rangle_{\text{OS}, \varepsilon} := \int \overline{F(\Theta A)} G(A) d\mu_{\varepsilon, \text{hg}}(A).$$

$$\langle F, F \rangle_{\text{OS}, \varepsilon} \geq 0.$$

$$\mathcal{N}_\varepsilon := \{F \in \mathfrak{A}_+ : \langle F, F \rangle_{\text{OS}, \varepsilon} = 0\}.$$

$$\mathcal{H}_{\varepsilon, +} := \overline{\mathfrak{A}_+ / \mathcal{N}_\varepsilon}^{\|\cdot\|_{\text{OS}, \varepsilon}}.$$

$$(S_u^\varepsilon F)(A) := F(\mathbb{T}_u^+ A), \quad u \geq 0.$$

$$\widehat{S}_u^\varepsilon[F] := [S_u^\varepsilon F].$$

$$\widehat{S}_u^\varepsilon = e^{-uH_\varepsilon}.$$

$$\mathcal{Q}_\varepsilon(F, F) := \lim_{u \downarrow 0} \frac{1}{u} \left(\langle F, F \rangle_{\text{OS}, \varepsilon} - \langle F, \widehat{S}_u^\varepsilon F \rangle_{\text{OS}, \varepsilon} \right).$$

$$= \lim_{u \downarrow 0} \frac{1}{u} \langle F, (I - \widehat{S}_u^\varepsilon) F \rangle_{\text{OS}, \varepsilon}.$$

$$\mathcal{Q}_\varepsilon(F, F) = \langle H_\varepsilon^{1/2} F, H_\varepsilon^{1/2} F \rangle_{\mathcal{H}_{\varepsilon, +}}.$$

$$\mathcal{Q}_\varepsilon(F, F) = \mathcal{Q}_{\varepsilon, E}(F, F) + \mathcal{Q}_{\varepsilon, B}(F, F).$$

$$\mathcal{Q}_{\varepsilon, E}(F, F) = \frac{g^2}{2} \int \left\langle F(A), \left(\int_{\Sigma_t} |\widehat{E}_\varepsilon(t, x)|^2 dx \right) F(A) \right\rangle d\mu_{\varepsilon, +}(A).$$

$$\mathcal{Q}_{\varepsilon, B}(F, F) = \frac{1}{2g^2} \int \left\langle F(A), \left(\int_{\Sigma_t} |\widehat{B}_\varepsilon(t, x)|^2 dx \right) F(A) \right\rangle d\mu_{\varepsilon, +}(A).$$

$$\mathcal{Q}_{\varepsilon, E}(F, F) \geq 0, \quad \mathcal{Q}_{\varepsilon, B}(F, F) \geq 0.$$

reflection-positive measure + positive-time transfer semigroup + infinitesimal generator

9. Cutoff-free physical charged sector.

$$\Omega_{\text{hg}} = \bigcup_{t \in (-T_*, T_*)} \{t\} \times \Sigma_t \subset \mathbb{R}^4.$$

$$\Sigma_t \subset \mathbb{R}^3 \quad \text{bounded Lipschitz.}$$

$$x_-(t), x_+(t) \in \Sigma_t, \quad |x_+(t) - x_-(t)| = R.$$

$$D_{R,t} = \bigcup_{s \in [0,R]} S_{t,s}.$$

$$S_{t,s} \cap S_{t,s'} = \emptyset \quad \text{for } s \neq s' \quad \text{up to null sets.}$$

$$|S_{t,s}| \leq A_* \quad \text{for a.e. } s \in [0, R].$$

$$\exists c_{\text{geo}} > 0 \quad \text{such that}$$

$$\int_{D_{R,t}} f(x) dx \geq c_{\text{geo}} \int_0^R \int_{S_{t,s}} f(x) dA(x) ds$$

$$\text{for every nonnegative measurable } f.$$

$$G \text{ compact connected Lie group,} \quad \mathfrak{g} = \text{Lie}(G).$$

$$\rho : G \rightarrow U(V_\rho) \quad \text{irreducible unitary representation,} \quad d\rho \neq 0.$$

$$d\rho : \mathfrak{g} \rightarrow \mathfrak{u}(V_\rho).$$

$$\{X^a\}_{a=1}^{\dim \mathfrak{g}} \quad \text{orthonormal with respect to an Ad-invariant inner product.}$$

$$\mathcal{K}_{R,\rho}(t) := V_\rho \otimes \mathcal{H}_{\text{gauge}}(t) \otimes V_\rho^*.$$

$$\mathcal{D}_t := V_\rho \otimes \mathcal{D}_{\text{gauge}}(t) \otimes V_\rho^* \subset \mathcal{K}_{R,\rho}(t).$$

$$\overline{\mathcal{D}_t} = \mathcal{K}_{R,\rho}(t).$$

$$\mathcal{G}_t := \text{Map}_{\text{adm}}(\Sigma_t, G).$$

$$h \in \mathcal{G}_t, \quad h_\pm := h(x_\pm(t)).$$

$$\mathbf{U}_t(h) : \mathcal{H}_{\text{gauge}}(t) \rightarrow \mathcal{H}_{\text{gauge}}(t) \quad \text{unitary.}$$

$$\mathbf{V}_{R,\rho,t}(h) := \rho(h_-) \otimes \mathbf{U}_t(h) \otimes \rho^*(h_+).$$

$$\mathbf{V}_{R,\rho,t}(h) \text{ is unitary on } \mathcal{K}_{R,\rho}(t) = V_\rho \otimes \mathcal{H}_{\text{gauge}}(t) \otimes V_\rho^*.$$

The physical charged sector is the gauge-equivariant subspace satisfying the gauge constraint.

$$\mathcal{K}_{R,\rho}^{\text{phys}}(t) := \{ \Psi \in \mathcal{K}_{R,\rho}(t) : \mathbf{V}_{R,\rho,t}(h)\Psi = \Psi \quad \forall h \in \mathcal{G}_t \}.$$

$$\mathcal{D}_t^{\text{phys}} := \mathcal{D}_t \cap \mathcal{K}_{R,\rho}^{\text{phys}}(t).$$

$$\overline{\mathcal{D}_t^{\text{phys}}} = \mathcal{K}_{R,\rho}^{\text{phys}}(t).$$

10. Casimir flux identity.

$$C_\rho := \sum_{a=1}^{\dim \mathfrak{g}} d\rho(X^a)^* d\rho(X^a).$$

$$d\rho(X^a)^* = -d\rho(X^a).$$

$$d\rho(X^a)^* d\rho(X^a) \geq 0.$$

$$C_\rho \geq 0.$$

$$\rho(g) d\rho(X^a) \rho(g)^{-1} = d\rho(\text{Ad}_g X^a).$$

$$\sum_a d\rho(\text{Ad}_g X^a)^* d\rho(\text{Ad}_g X^a) = \sum_a d\rho(X^a)^* d\rho(X^a).$$

$$\rho(g) C_\rho \rho(g)^{-1} = C_\rho.$$

$$C_\rho \rho(g) = \rho(g) C_\rho \quad \forall g \in G.$$

$$\rho \text{ irreducible} \implies C_\rho = C_2(\rho) I_{V_\rho}.$$

$$C_2(\rho) \geq 0.$$

$$C_2(\rho) = 0 \implies \sum_a \|d\rho(X^a)v\|^2 = 0 \quad \forall v \in V_\rho.$$

$$\implies d\rho(X^a)v = 0 \quad \forall a, \forall v \in V_\rho.$$

$$\implies d\rho(X^a) = 0 \quad \forall a.$$

$$\implies d\rho = 0.$$

$$d\rho \neq 0 \implies C_2(\rho) > 0.$$

$$\begin{aligned} \sum_{a=1}^{\dim \mathfrak{g}} \|(d\rho(X^a) \otimes I \otimes I)\Psi\|^2 &= \left\langle \Psi, \left(\sum_a d\rho(X^a)^* d\rho(X^a) \otimes I \otimes I \right) \Psi \right\rangle. \\ &= \langle \Psi, (C_2(\rho) I \otimes I \otimes I) \Psi \rangle. \end{aligned}$$

$$= C_2(\rho) \|\Psi\|^2.$$

The orientation of $S_{t,s}$ is chosen so that the enclosed endpoint charge is the left V_ρ factor.

With this convention, the Gauss-law flux through $S_{t,s}$ acts as $d\rho(X^a) \otimes I \otimes I$.

If the opposite orientation is chosen, the sign changes, but the Casimir norm-square estimate is unchanged.

$$\widehat{\mathcal{J}}_t^a(s) : \mathcal{D}_t \rightarrow \mathcal{K}_{R,\rho}(t).$$

$$\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi = (d\rho(X^a) \otimes I \otimes I)\Psi \quad (\Psi \in \mathcal{D}_{\Lambda,t}^{\text{phys}})$$

for a.e. $s \in [0, R]$.

$$\begin{aligned} \sum_{a=1}^{\dim \mathfrak{g}} \|\widehat{\mathcal{J}}_t^a(s)\Psi\|^2 &\leq |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t, x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle. \\ \sum_a \|\widehat{\mathcal{J}}_t^a(s)\Psi\|^2 &= \sum_a \|(d\rho(X^a) \otimes I \otimes I)\Psi\|^2. \\ &= C_2(\rho) \|\Psi\|^2. \end{aligned}$$

$$\begin{aligned} C_2(\rho) \|\Psi\|^2 &\leq |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t, x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle. \\ \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t, x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle &\geq \frac{C_2(\rho)}{|S_{t,s}|} \|\Psi\|^2. \\ |S_{t,s}| &\leq A_*. \end{aligned}$$

$$\begin{aligned} \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t, x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle &\geq \frac{C_2(\rho)}{A_*} \|\Psi\|^2. \\ |\widehat{E}(t, x)|^2 &\geq |\widehat{E}(t, x) \cdot \nu_{t,s}(x)|^2. \end{aligned}$$

$$\begin{aligned} \left\langle \Psi, \left(\int_{D_{R,t}} |\widehat{E}(t, x)|^2 dx \right) \Psi \right\rangle &\geq c_{\text{geo}} \int_0^R \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}(t, x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle ds. \\ &\geq c_{\text{geo}} \int_0^R \frac{C_2(\rho)}{A_*} \|\Psi\|^2 ds. \\ &= \frac{c_{\text{geo}} C_2(\rho)}{A_*} R \|\Psi\|^2. \end{aligned}$$

11. Finite-rank gauge-compatible cutoff.

$$\Lambda > 0.$$

$$P_\Lambda : \mathcal{H}_{\text{gauge}}(t) \rightarrow \mathcal{H}_{\text{gauge}}(t) \quad \text{finite-rank orthogonal projection.}$$

$$P_\Lambda \rightarrow I \quad \text{strongly as } \Lambda \rightarrow \infty.$$

$$P_\Lambda^{R,\rho} := I_{V_\rho} \otimes P_\Lambda \otimes I_{V_\rho^*}.$$

$$P_\Lambda^{R,\rho} \rightarrow I \quad \text{strongly on } \mathcal{K}_{R,\rho}(t).$$

$$P_\Lambda \mathbf{U}_t(h) = \mathbf{U}_t(h) P_\Lambda \quad \forall h \in \mathcal{G}_t.$$

$$P_\Lambda^{R,\rho} = I_{V_\rho} \otimes P_\Lambda \otimes I_{V_\rho^*}.$$

$$P_\Lambda^{R,\rho} \mathbf{V}_{R,\rho,t}(h) = \mathbf{V}_{R,\rho,t}(h) P_\Lambda^{R,\rho}.$$

$$\Psi \in \mathcal{K}_{R,\rho}^{\text{phys}}(t) \implies P_\Lambda^{R,\rho} \Psi \in \mathcal{K}_{R,\rho}^{\text{phys}}(t).$$

$$\mathcal{D}_{\Lambda,t}^{\text{phys}} := P_\Lambda^{R,\rho} \mathcal{D}_t^{\text{phys}}.$$

$$\overline{\mathcal{D}_{\Lambda,t}^{\text{phys}}} = P_\Lambda^{R,\rho} \mathcal{K}_{R,\rho}^{\text{phys}}(t).$$

$$\dim \mathcal{D}_{\Lambda,t}^{\text{phys}} < \infty.$$

$$\widehat{E}_\Lambda(t, x) := P_\Lambda^{R,\rho} \widehat{E}(t, x) P_\Lambda^{R,\rho}.$$

$$\widehat{B}_\Lambda(t, x) := P_\Lambda^{R,\rho} \widehat{B}(t, x) P_\Lambda^{R,\rho}.$$

$$\widehat{\mathcal{J}}_{\Lambda,t}^a(s) := P_\Lambda^{R,\rho} \widehat{\mathcal{J}}_t^a(s) P_\Lambda^{R,\rho}.$$

$$P_\Lambda^{R,\rho} (d\rho(X^a) \otimes I \otimes I) = (d\rho(X^a) \otimes I \otimes I) P_\Lambda^{R,\rho}.$$

$$\widehat{\mathcal{J}}_{\Lambda,t}^a(s) \Psi = (d\rho(X^a) \otimes I \otimes I) \Psi \quad (\Psi \in \mathcal{D}_{\Lambda,t})$$

$$\text{for a.e. } s \in [0, R].$$

$$\begin{aligned} \sum_a \|\widehat{\mathcal{J}}_{\Lambda,t}^a(s) \Psi\|^2 &= \sum_a \|(d\rho(X^a) \otimes I \otimes I) \Psi\|^2 \\ &= C_2(\rho) \|\Psi\|^2. \end{aligned}$$

$$\begin{aligned}
\sum_a \|\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi\|^2 &\leq |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle. \\
C_2(\rho)\|\Psi\|^2 &\leq |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle. \\
\left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle &\geq \frac{C_2(\rho)}{A_*} \|\Psi\|^2. \\
|\widehat{E}_{\Lambda}(t,x)|^2 &\geq |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2. \\
\left\langle \Psi, \left(\int_{D_{R,t}} |\widehat{E}_{\Lambda}(t,x)|^2 dx \right) \Psi \right\rangle &\geq c_{\text{geo}} \int_0^R \frac{C_2(\rho)}{A_*} \|\Psi\|^2 ds. \\
&= \frac{c_{\text{geo}} C_2(\rho)}{A_*} R \|\Psi\|^2.
\end{aligned}$$

In this primitive charged-sector formulation, the external color sources are not inserted as point-supported distributions in the gauge field. They are implemented only as finite-dimensional endpoint representation factors. Therefore no singular local source self-energy counterterm is part of the Yang–Mills quadratic form.

$$\mathfrak{r}_{\Lambda,t} := 0.$$

$$\mathfrak{r}_{\Lambda,t}(\Psi, \Psi) = 0 \quad \forall \Psi \in \mathscr{D}_{\Lambda,t}^{\text{phys}}.$$

$$\mathfrak{r}_{\Lambda,t}(\Psi, \Psi) \geq 0.$$

$$\kappa_{\Lambda} := 0.$$

$$\kappa := \sup_{\Lambda} \kappa_{\Lambda} = 0.$$

$$\mathfrak{q}_{\Lambda,R,\rho,t}^{\text{phys}} : \mathscr{D}_{\Lambda,t}^{\text{phys}} \times \mathscr{D}_{\Lambda,t}^{\text{phys}} \rightarrow \mathbb{C}.$$

$$\mathfrak{q}_{\Lambda,R,\rho,t}^{\text{phys}}(\Psi, \Psi) := \frac{g^2}{2} \left\langle \Psi, \left(\int_{\Sigma_t} |\widehat{E}_{\Lambda}(t,x)|^2 dx \right) \Psi \right\rangle + \frac{1}{2g^2} \left\langle \Psi, \left(\int_{\Sigma_t} |\widehat{B}_{\Lambda}(t,x)|^2 dx \right) \Psi \right\rangle.$$

$$\left\langle \Psi, \left(\int_{\Sigma_t} |\widehat{E}_{\Lambda}(t,x)|^2 dx \right) \Psi \right\rangle \geq 0.$$

$$\left\langle \Psi, \left(\int_{\Sigma_t} |\widehat{B}_{\Lambda}(t,x)|^2 dx \right) \Psi \right\rangle \geq 0.$$

$$\mathfrak{q}_{\Lambda,R,\rho,t}^{\text{phys}}(\Psi, \Psi) \geq 0.$$

$$\mathfrak{q}_{\Lambda,R,\rho,t}^{\text{phys}} \text{ is lower semibounded on } \mathscr{D}_{\Lambda,t}^{\text{phys}}.$$

$$\dim \mathcal{D}_{\Lambda,t}^{\text{phys}} < \infty \implies \mathfrak{q}_{\Lambda,R,\rho,t}^{\text{phys}} \text{ is closed on } P_{\Lambda}^{R,\rho} \mathcal{K}_{R,\rho}^{\text{phys}}(t).$$

$$\exists! H_{\Lambda,R,\rho}^{\text{phys}}(t) = H_{\Lambda,R,\rho}^{\text{phys}}(t)^* \quad \text{on } P_{\Lambda}^{R,\rho} \mathcal{K}_{R,\rho}^{\text{phys}}(t).$$

$$H_{\Lambda,R,\rho}^{\text{phys}}(t) \quad \text{is associated with } \mathfrak{q}_{\Lambda,R,\rho,t}^{\text{phys}}.$$

12. Linear lower bound at finite cutoff.

$$\mathfrak{q}_{\Lambda,R,\rho,t}^{\text{phys}} = \mathcal{Q}_{\Lambda}^{R,\rho} \quad \text{on the cutoff physical charged sector.}$$

$$\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi = (d\rho(X^a) \otimes I \otimes I)\Psi \quad \Psi \in \mathcal{D}_{\Lambda,t}^{\text{phys}}.$$

$$\begin{aligned} \sum_a \|\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi\|^2 &= \sum_a \|(d\rho(X^a) \otimes I \otimes I)\Psi\|^2 \\ &= C_2(\rho)\|\Psi\|^2. \end{aligned}$$

$$\sum_a \|\widehat{\mathcal{J}}_{\Lambda,t}^a(s)\Psi\|^2 \leq |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle.$$

$$C_2(\rho)\|\Psi\|^2 \leq |S_{t,s}| \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle.$$

$$|S_{t,s}| \leq A_*.$$

$$\left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle \geq \frac{C_2(\rho)}{A_*} \|\Psi\|^2.$$

$$|\widehat{E}_{\Lambda}(t,x)|^2 \geq |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2.$$

$$\left\langle \Psi, \left(\int_{D_{R,t}} |\widehat{E}_{\Lambda}(t,x)|^2 dx \right) \Psi \right\rangle \geq c_{\text{geo}} \int_0^R \left\langle \Psi, \left(\int_{S_{t,s}} |\widehat{E}_{\Lambda}(t,x) \cdot \nu_{t,s}(x)|^2 dA(x) \right) \Psi \right\rangle ds.$$

$$\geq c_{\text{geo}} \int_0^R \frac{C_2(\rho)}{A_*} \|\Psi\|^2 ds.$$

$$= \frac{c_{\text{geo}} C_2(\rho)}{A_*} R \|\Psi\|^2.$$

$$\mathfrak{q}_{\Lambda,R,\rho,t}^{\text{phys}}(\Psi, \Psi) \geq \frac{g^2}{2} \left\langle \Psi, \left(\int_{D_{R,t}} |\widehat{E}_{\Lambda}(t,x)|^2 dx \right) \Psi \right\rangle.$$

$$\geq \frac{g^2}{2} \frac{c_{\text{geo}} C_2(\rho)}{A_*} R \|\Psi\|^2.$$

$$\sigma := \frac{g^2 c_{\text{geo}} C_2(\rho)}{2A_*}.$$

$$\mathfrak{q}_{\Lambda, R, \rho, t}^{\text{phys}}(\Psi, \Psi) \geq \sigma R \|\Psi\|^2.$$

$$H_{\Lambda, R, \rho}^{\text{phys}}(t) \geq \sigma R I.$$

$$E_{0, \Lambda}^{\text{phys}}(R, t) := \inf \sigma \left(H_{\Lambda, R, \rho}^{\text{phys}}(t) \right).$$

$$E_{0, \Lambda}^{\text{phys}}(R, t) \geq \sigma R.$$

$$\boxed{E_{0, \Lambda}^{\text{phys}}(R, t) \geq \frac{g^2 c_{\text{geo}} C_2(\rho)}{2A_*} R.}$$

$$\kappa_{\Lambda} = 0 \quad \forall \Lambda.$$

$$\sup_{\Lambda} \kappa_{\Lambda} = 0.$$

$$\kappa = 0.$$

13. Cutoff removal.

$$\mathfrak{q}_{\Lambda, R, \rho, t}^{\text{phys}} \xrightarrow{\text{Mosco}} \mathfrak{q}_{R, \rho, t}^{\text{phys}}.$$

$$\Psi_{\Lambda} \rightharpoonup \Psi \implies \mathfrak{q}_{R, \rho, t}^{\text{phys}}(\Psi, \Psi) \leq \liminf_{\Lambda \rightarrow \infty} \mathfrak{q}_{\Lambda, R, \rho, t}^{\text{phys}}(\Psi_{\Lambda}, \Psi_{\Lambda}).$$

$$\forall \Psi \in \text{Dom} \left(\mathfrak{q}_{R, \rho, t}^{\text{phys}} \right) \quad \exists \Psi_{\Lambda} \rightarrow \Psi$$

$$\text{such that} \quad \mathfrak{q}_{R, \rho, t}^{\text{phys}}(\Psi, \Psi) = \lim_{\Lambda \rightarrow \infty} \mathfrak{q}_{\Lambda, R, \rho, t}^{\text{phys}}(\Psi_{\Lambda}, \Psi_{\Lambda}).$$

$$\mathfrak{q}_{R, \rho, t}^{\text{phys}} \text{ is closed and lower semibounded.}$$

$$H_{\Lambda, R, \rho}^{\text{phys}}(t) \rightarrow H_{R, \rho}^{\text{phys}}(t) \text{ in strong resolvent sense.}$$

$$\Psi \in \text{Dom} \left(\mathfrak{q}_{R, \rho, t}^{\text{phys}} \right).$$

$$\exists \Psi_{\Lambda} \in \mathcal{D}_{\Lambda, t}^{\text{phys}}, \quad \Psi_{\Lambda} \rightarrow \Psi.$$

$$\mathfrak{q}_{R, \rho, t}^{\text{phys}}(\Psi, \Psi) = \lim_{\Lambda \rightarrow \infty} \mathfrak{q}_{\Lambda, R, \rho, t}^{\text{phys}}(\Psi_{\Lambda}, \Psi_{\Lambda}).$$

$$\mathfrak{q}_{\Lambda, R, \rho, t}^{\text{phys}}(\Psi_{\Lambda}, \Psi_{\Lambda}) \geq \sigma R \|\Psi_{\Lambda}\|^2.$$

$$\lim_{\Lambda \rightarrow \infty} \mathfrak{q}_{\Lambda, R, \rho, t}^{\text{phys}}(\Psi_\Lambda, \Psi_\Lambda) \geq \lim_{\Lambda \rightarrow \infty} \sigma R \|\Psi_\Lambda\|^2.$$

$$\mathfrak{q}_{R, \rho, t}^{\text{phys}}(\Psi, \Psi) \geq \sigma R \|\Psi\|^2.$$

$$H_{R, \rho}^{\text{phys}}(t) \geq \sigma R I.$$

$$E_0^{\text{phys}}(R, t) := \inf \sigma \left(H_{R, \rho}^{\text{phys}}(t) \right).$$

$$E_0^{\text{phys}}(R, t) \geq \sigma R.$$

$$E_0^{\text{phys}}(R, t) \geq \frac{g^2 c_{\text{geo}} C_2(\rho)}{2A_*} R.$$

14. Evolution estimate.

Assume that the non-autonomous equation

$$\partial_t \Psi(t) = -H_{R, \rho}^{\text{phys}}(t) \Psi(t)$$

generates a contraction evolution family $U_{R, \rho}^{\text{phys}}(t, s)$.

$$H_{R, \rho}^{\text{phys}}(t) \geq \sigma R I.$$

$$\frac{d}{dt} \|\Psi(t)\|^2 = -2 \left\langle \Psi(t), H_{R, \rho}^{\text{phys}}(t) \Psi(t) \right\rangle.$$

$$\leq -2\sigma R \|\Psi(t)\|^2.$$

$$\frac{d}{dt} \left(e^{2(t-s)\sigma R} \|\Psi(t)\|^2 \right) \leq 0.$$

$$e^{2(t-s)\sigma R} \|\Psi(t)\|^2 \leq \|\Psi(s)\|^2.$$

$$\|\Psi(t)\|^2 \leq e^{-2(t-s)\sigma R} \|\Psi(s)\|^2.$$

$$\|\Psi(t)\| \leq e^{-(t-s)\sigma R} \|\Psi(s)\|.$$

$$\|U_{R, \rho}^{\text{phys}}(t, s)\| \leq e^{-(t-s)\sigma R}.$$